# Tactix: matched diagnosis of the lost WV100 solve

Tactix Development Project

October 3, 2026

The first matched comparison on `snw_16_9_CCSpre` completes with independently verified SAT models for predecessor `f6f01982` and repaired source `375b621b`. The candidate takes 1,035.187 seconds versus 1,046.445 for the predecessor, with identical complete printed work and certificate hash. **The original WV100 gate remains NOT ACCEPTED.** This pair does not reproduce a candidate slowdown or explain the original timeout; it does not replace the full gate.

## Matched measurements

SLURM 1747087 is COMPLETED, exit 0, elapsed 34:50. The sequential predecessor/current runs use the same 12-logical-CPU/32-GiB allocation on node017, the same affinity/solver CPU, immutable input/arguments and original 1,200-second solver cap. Both original-formula model checks verify without timeout. Twenty pin bindings verify before and after execution and currently.

| Measurement | Predecessor | Candidate |
|---|---:|---:|
| Solver wall seconds | 1,046.445 | 1,035.187 |
| Inprocessing slot seconds | 253.048 | 250.897 |
| Search/scaffolding seconds | 793.018 | 783.945 |
| Vivify seconds | 147.682 | 145.508 |
| Instructions | 5,178,110,235,247 | 5,178,103,124,802 |
| Cycles | 3,974,276,563,543 | 3,933,400,938,042 |
| Page faults | 432,795 | 432,758 |

Both record 3,936,628 decisions, 1,220,557 conflicts, 7,077,999,954 propagations, 35 inprocessing phases and 39,296 restarts. Candidate time is 1.076% lower; instructions differ by about 0.000137%, cycles by -1.029% and page faults by -37. This does not support a large candidate work or instruction-cost increase in this pair. Certificate equality is correspondence evidence; independent model checks establish these SAT results.

The original timeout lacks final search and hardware counters. Order, cache and occupancy effects remain possible, and SAT model bytes are not compared. The observed pair therefore does not establish the original slowdown's cause. Other three default gates and scoped feedback measurement remain accepted, as described in the terminal gate report.

## Bounded follow-up and acceptance

Before submission, the protocol registers exactly one reverse pair and one complete WV100 recovery campaign. Both are now live:

- Reverse pair 1747088, watcher 40486: candidate first, predecessor second, unchanged pins, input, caps and oracles, to expose order sensitivity and repeat spread. Timeout, work divergence or material candidate slowdown calls for diagnosis rather than another automatic repeat.

- Recovery array 1747089, watcher 10695: all original 100 WV100 instances, unchanged accepted baseline, resources, solver/checker caps and retention/censoring contracts. The maintained scorer must accept the complete gate; no missing row or checker rejection is allowed.

No automatic retry loop, tuning, throttle, node constraint, allowance increase or isolated-row substitution is permitted. The original failed campaign is preserved. Each proof's complete pipeline still has 1,800 seconds; comparable baseline time $t$ retains the $t + \max(0.30t, 20\,\mathrm{s})$ bound. Censored proofs remain unverified. Preflight verifies seventeen recovery and twenty reverse bindings, with no solver execution during preparation.

Full Debug 1746924 and Claude-owned fresh/targeted fuzz remain live. All 795 source bindings are retained; no private campaign contents were inspected. Next: collect the two bounded follow-ups and exact Debug/fuzz acceptance, then audit the complete original objective. No source, reward, learning, cleanup, unrelated cancellation or push change occurred. Whole-goal acceptance is still pending.

The technical report and hashed companion retain both raw rows, complete analysis, immutable provenance, terminal metadata and the preregistration/submissions.